EXPERIMENTS

1. EXPERIMENT NO: **1**

2. TITLE: **PERFORMANCE OF POINT-TO-POINT NETWORK**

3. LEARNING OBJECTIVES:

- To simulate point to point network
- Understanding working of duplex links
- Analyze network performance.

4. AIM: To implement a three nodes point–to– point network with duplex links between them. Set the queue size and vary the bandwidth and find the number of packets dropped.

5. THEORY:

Point-to-Point (P2P) Network: A point-to-point network is a topology where direct links are established between two nodes. In a three-node point-to-point duplex network, each node is connected to another using bidirectional (duplex) links, allowing two-way communication.

Duplex Links in NS2 : A duplex link in NS2 is created using the duplex-link command, which connects two nodes in both directions. Each link can be configured with:

- Bandwidth (e.g., 1Mb)
- Propagation delay (e.g., 10ms)
- Queue type (e.g., DropTail or RED)
- Queue size (e.g., 50 packets)

Traffic Generation and Packet Flow: To simulate communication:

- A **TCP Agent** is attached to the source node (e.g., n0).
- A **TCPSink Agent** is attached to the destination node (e.g., n2).
- An **FTP application** is attached to the TCP agent to generate continuous traffic.

Traffic flows from the source to destination across the duplex links (n0 → n1 → n2), passing through intermediate routers (nodes).

Queue Management and Packet Drops: Each duplex link includes a buffer (queue) that holds packets during congestion. If the queue becomes full, incoming packets are dropped.

By:

- Fixing the queue size, and
- Varying the bandwidth (e.g., 0.5 Mbps, 1 Mbps, 2 Mbps),we can analyse how link capacity impacts packet drops.

Trace File Analysis: NS2 generates a **trace file (.tr)** that logs all events like:

- Packet enqueue (+)
- Packet dequeue (-)
- Packet drop (d)
- Packet receive (r)

By parsing this file (using **AWK** or **Python scripts**), the **number of dropped packets** can be counted and correlated

6. PROCEDURE :

Step 1: Open NSG2.1

```
ailab@ailab-H510M-H-V2: ~/Nagaraja/java -jar NSG2.1.jar
```

Step 2: Create Topology (Following steps specified in theory)

Step 3: Set parameters Nam file name : file_name.nam

Trace file name : file_name.tr

Step 4: Open tcl program and save as program_name.tcl

Step 5: Edit program for varying Band width , Packet Size etc

Step 6: Run Simulation

```
ailab@ailab-H510M-H-V2:~/Nagaraja/cnlab/program1$ ns p1.tcl
```

Step 7: Edit awk program file to count Number of packets Received /Dropped and display result

```
ailab@ailab-H510M-H-V2:~/Nagaraja/cnlab/program1$ gedit p1.awk
```

Step 8 : Run awk file and observe results.

```
ailab@ailab-H510M-H-V2:~/Nagaraja/cnlab/program1$ awk -f p1.awk p1.tr
```

7. PROGRAMS:

p1.tcl

```
#===================================
#     Simulation parameters setup
#===================================
set val(stop)   10.0 ; # time of simulation end
#===================================
#        Initialization
#===================================
#Create a ns simulator
set ns [new Simulator]
#Open the NS trace file
set tracefile [open p1.tr w]
$ns trace-all $tracefile

#Open the NAM trace file
set namfile [open p1.nam w]
$ns namtrace-all $namfile
#===================================
#        Nodes Definition
#===================================
#Create 3 nodes
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]

#===================================
#        Links Definition
#===================================
```

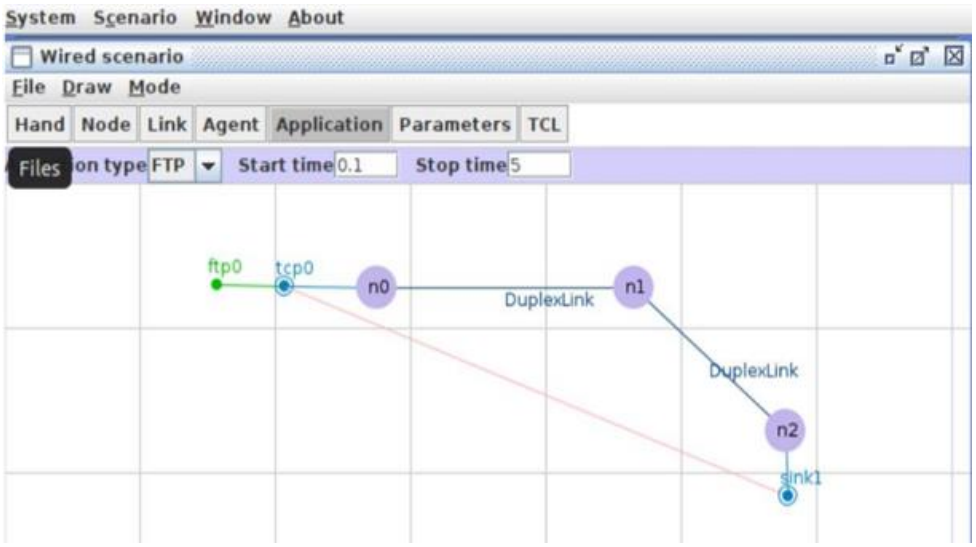
```
#Create links between nodes
$ns duplex-link $n0 $n1 100.0Mb 10ms DropTail
$ns queue-limit $n0 $n1 50
$ns duplex-link $n1 $n2 100.0Mb 10ms DropTail
$ns queue-limit $n1 $n2 50

#Give node position (for NAM)
$ns duplex-link-op $n0 $n1 orient right
$ns duplex-link-op $n1 $n2 orient right-down
#===================================
#        Agents Definition
#===================================
#Setup a TCP connection
set tcp0 [new Agent/TCP]
$ns attach-agent $n0 $tcp0
set sink1 [new Agent/TCPSink]
$ns attach-agent $n2 $sink1
$ns connect $tcp0 $sink1
$tcp0 set packetSize_ 1500

#===================================
#        Applications Definition
#===================================
#Setup a FTP Application over TCP connection

set ftp0 [new Application/FTP]
$ftp0 attach-agent $tcp0
$ns at 1.0 "$ftp0 start"
$ns at 15.0 "$ftp0 stop"
#===================================
#        Termination
#===================================
#Define a 'finish' procedure
proc finish {} {
    global ns tracefile namfile
    $ns flush-trace
    close $tracefile
    close $namfile
    exec nam p1.nam &
    exit 0
}
$ns at $val(stop) "$ns nam-end-wireless $val(stop)"
$ns at $val(stop) "finish"
$ns at $val(stop) "puts \"done\" ; $ns halt"
$ns run
```

pcktdroprecv.awk

```
BEGIN{
count=0;
a=0;
b=0;
}
{
if($1=="d")
{
count++;
}
if($1=="r" &&$3=="1"&&$4=="2"&&$5=="tcp"&&$6=="1540")
{
a++;
}
if($1=="r" &&$3=="0"&&$4=="1"&&$5=="tcp"&&$6=="1540")
{
b++;
}
}
END{
printf("No of Dropped packet is %d\n", count);
printf("No. of packets recieved: 1-->2 %d\n", a);
printf("No. of packets recieved: 0-->1 %d\n", b);
}
```

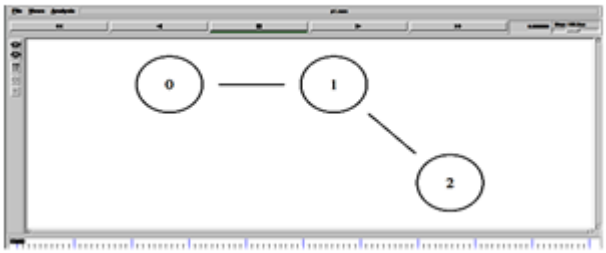
8. RESULTS & CONCLUSIONS:

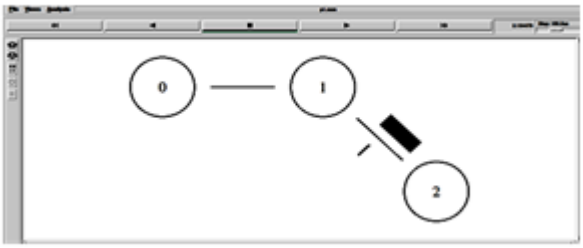
Topology for point to point connection using NSG2.1



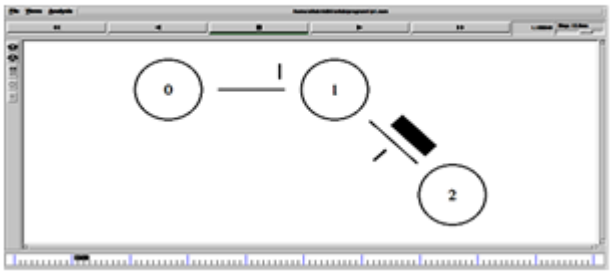
Network Simulator

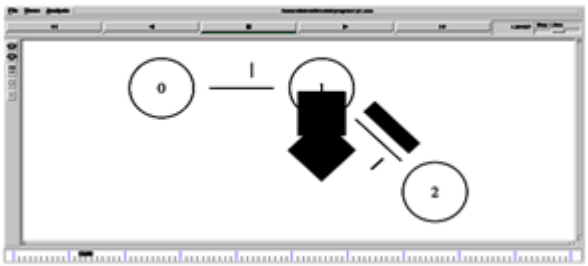
Case 1 No Packet drop



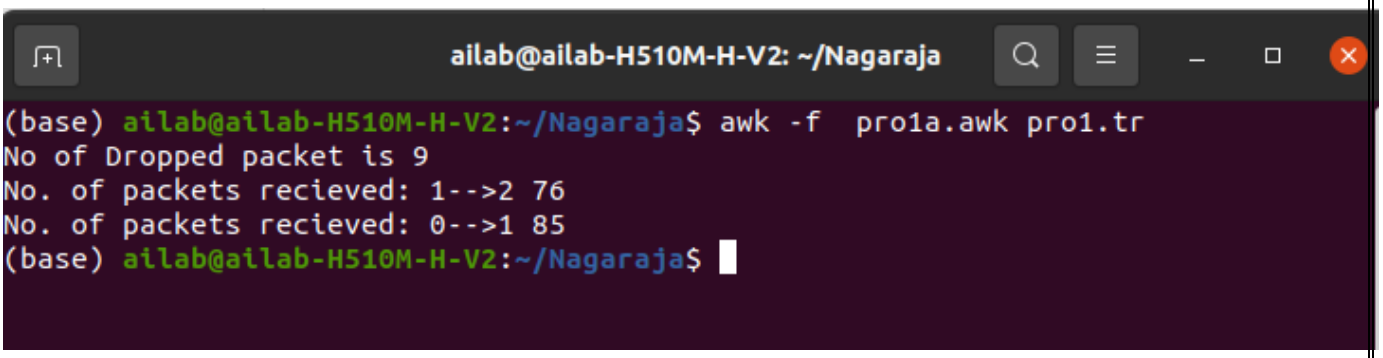


Case 2 : Packet drop





2)No. Of Packets Dropped & Received



Note : For each case changes made bandwidth, packet size link definition portion of **tcl** file and corresponding result are to be recorded.

9. LEARNING OUTCOME

- Knowledge of bandwidth change by the change of queue size is analyzed.
- Analysis of the performance is carried out.

10. APPLICATION AREAS:

- Internet/Intranet configurations.
- ISP access networks (bridged or routed).
- LAN to LAN applications (bridge or router).

11. REMARKS: